

STRUCTURED BILINEAR DIRECTIVITY CONTROL FOR MULTI-CHANNEL PARAMETRIC ARRAY LOUDSPEAKERS

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ABSTRACT

Directivity control of a multi-channel parametric array loudspeaker requires accounting for the nonlinear coupling between different channels. For an N -channel array, all carrier-sideband combinations produce N^2 channel-pair responses, making accurate model construction computationally demanding. This paper formulates the far-field audio directivity as a bilinear function of the carrier and sideband weights. For identical and uniformly spaced channels, each pair response separates into an absolute-position phase and a relative-displacement kernel, revealing a diagonal-Toeplitz structure. Angular mirror symmetry and the decay of distant-pair contributions further reduce the number of independently evaluated responses from N^2 to $L + 1$ ($L < N$). For the considered 24-channel array, only five kernels are required instead of 576 pair responses while retaining 99.84% of the matrix energy. The reduced model closely reproduces the full-model directivity and preserves the performance of joint carrier-sideband pressure matching, which improves the suppression-region level by 5.7 dB over conventional steering.

Index Terms— parametric array loudspeaker, directivity control, bilinear model, Toeplitz structure, pressure matching

1. INTRODUCTION

Parametric array loudspeakers (PALs) generate highly directional audible sound through the nonlinear interaction of ultrasonic waves [1–4]. Dividing the radiating aperture into independently driven channels enables electronic beam steering [5–9] and supports applications such as multibeam reproduction [10] and active noise control [11]. Unlike a conventional loudspeaker array, however, a multi-channel PAL does not produce the audible field through a linear superposition of channel responses. Instead, the carrier and sideband fields form a distributed virtual source, and every carrier-sideband channel pair contributes a nonlinear interaction to the resulting directivity [12, 13]. Consequently, an N -channel PAL generally involves N^2 direction-dependent pair responses, which makes accurate model construction a central challenge for directivity control.

Existing models address different parts of this challenge. Convolution directivity models provide efficient far-field predictions by representing self-demodulation through analytical convolution [14, 15], and have facilitated beam-pattern analysis and optimization [16–18]. Their simplified formulations, however, do not explicitly resolve the intricate spatial distribution of the nonlinear interaction, which can limit their accuracy for finite, discrete, and steered arrays. Time-domain and virtual-source methods retain this distributed interaction more directly [19, 20], whereas k -space methods enable efficient numerical propagation for complex array configurations [21]. Although these methods improve the prediction of nonlinear sound fields, constructing a complete multi-channel control model still requires evaluating all N^2 channel-pair responses. The internal structure of the resulting nonlinear coupling matrix, particularly its redundancy and effective sparsity, has not been systematically exploited.

To address this gap, this paper derives a bilinear directivity model and analyzes the structure of its nonlinear interaction matrix. Translation invariance, angular mirror symmetry, and separation-dependent coupling decay reduce the required independent responses from N^2 to $L + 1$. The reduced representation is then combined with alternating pressure matching for joint carrier-sideband control. Numerical results verify its reconstruction accuracy and show that it preserves the control performance of the full model.

2. BILINEAR DIRECTIVITY MODEL

2.1. Multi-channel PAL directivity

Consider an N -channel PAL driven by a carrier at angular frequency ω_c and an upper sideband at $\omega_s = \omega_c + \omega_a$. Let $p_{\nu,n}(\mathbf{r}_\nu, \omega_\nu)$ denote the ultrasonic pressure generated by the n th channel under unit excitation, where $\nu \in \{c, s\}$. The corresponding multi-channel transfer vector is

$$\mathbf{p}_\nu(\mathbf{r}_\nu, \omega_\nu) = [p_{\nu,1}(\mathbf{r}_\nu, \omega_\nu) \cdots p_{\nu,N}(\mathbf{r}_\nu, \omega_\nu)]^T. \quad (1)$$

For a channel-weight vector $\mathbf{w}_\nu$, the total ultrasonic pressure is

$$p_\nu(\mathbf{r}_\nu, \omega_\nu) = \mathbf{p}_\nu^T(\mathbf{r}_\nu, \omega_\nu) \mathbf{w}_\nu. \quad (2)$$

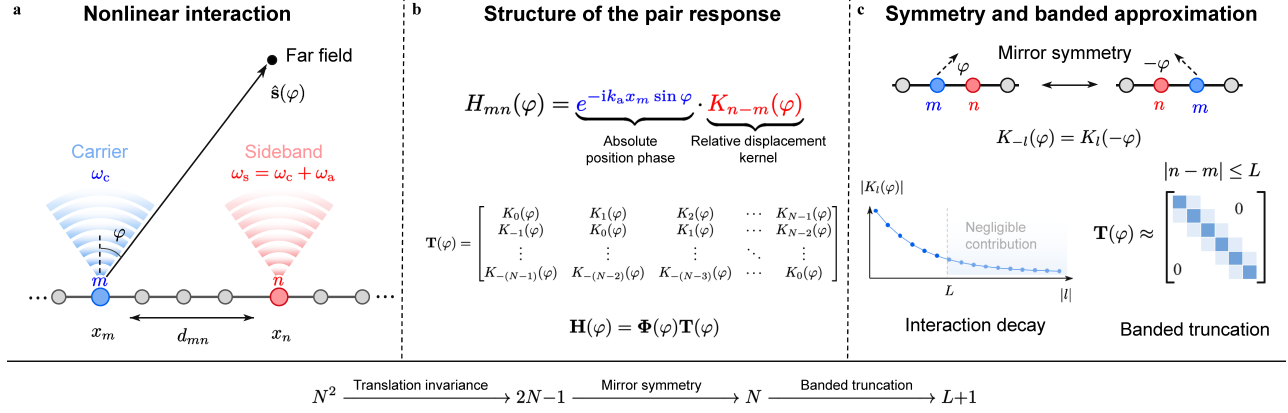


Fig. 1. Bilinear channel interaction and the resulting structured directivity model: (a) nonlinear interaction; (b) diagonal-Toeplitz structure; (c) mirror symmetry and banded approximation.

Under the quasilinear approximation [1, 2], the virtual source at the positive difference frequency ω_a is

$$q_a(\mathbf{r}_v, \omega_a) = \frac{\beta \omega_a}{i \rho_0^2 c_0^4} p_c^*(\mathbf{r}_v, \omega_c) p_s(\mathbf{r}_v, \omega_s). \quad (3)$$

Substituting (2) into (3) gives

$$q_a(\mathbf{r}_v, \omega_a) = \frac{\beta \omega_a}{i \rho_0^2 c_0^4} \mathbf{w}_c^H \mathbf{p}_c^*(\mathbf{r}_v, \omega_c) \mathbf{p}_s^T(\mathbf{r}_v, \omega_s) \mathbf{w}_s. \quad (4)$$

The outer product $\mathbf{p}_c^* \mathbf{p}_s^T$ contains all carrier-sideband channel nonlinear interactions. Its (m, n) th entry is the local interaction between carrier channel m and sideband channel n .

The observation direction is parameterized as $\hat{\mathbf{s}}(\theta_0, \varphi) = [\sin \varphi \cos \theta_0 \quad \sin \varphi \sin \theta_0 \quad \cos \varphi]^T$. This work considers the xOz plane for simplicity, for which $\theta_0 = 0$ and $\hat{\mathbf{s}}(\varphi) = [\sin \varphi, 0, \cos \varphi]^T$. With $k_a = \omega_a/c_0$, the far-field audio directivity is

$$D_a(\varphi, \omega_a) = -\frac{i \rho_0 \omega_a}{4\pi} \iiint_{\mathbb{R}^3} q_a(\mathbf{r}_v, \omega_a) \times e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_v} d^3 \mathbf{r}_v. \quad (5)$$

Substituting (4) into (5) and moving the excitation weights outside the integral yields the bilinear directivity model

$$D_a(\varphi, \omega_a) = \mathbf{w}_c^H \mathbf{H}(\varphi; \omega_c, \omega_s) \mathbf{w}_s, \quad (6)$$

where $\mathbf{H}(\varphi; \omega_c, \omega_s)$ is the nonlinear interaction matrix, defined as

$$\mathbf{H}(\varphi; \omega_c, \omega_s) = -\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iiint_{\mathbb{R}^3} \mathbf{p}_c^*(\mathbf{r}_v, \omega_c) \times \mathbf{p}_s^T(\mathbf{r}_v, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_v} d^3 \mathbf{r}_v. \quad (7)$$

The (m, n) th entry of (7) is

$$H_{mn}(\varphi) = -\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iiint_{\mathbb{R}^3} p_{c,m}^*(\mathbf{r}_v, \omega_c) \times p_{s,n}(\mathbf{r}_v, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_v} d^3 \mathbf{r}_v. \quad (8)$$

Therefore, $H_{mn}(\varphi)$ is the complex far-field response generated by one carrier-sideband channel pair. The matrix $\mathbf{H}(\varphi)$ contains the ultrasonic propagation and nonlinear interaction physics, whereas $\mathbf{w}_c$ and $\mathbf{w}_s$ determine how the pairwise responses are combined.

2.2. Displacement-kernel structure

Assume that the channels are identical and uniformly arranged along the x axis, with $x_n = (n-1)d$. In a homogeneous propagation environment, the channel fields satisfy

$$p_{\nu,n}(\mathbf{r}_v, \omega_\nu) = p_{\nu,0}(\mathbf{r}_v - x_n \mathbf{e}_x, \omega_\nu), \quad (9)$$

where $p_{\nu,0}$ denotes the field generated by a reference channel located at the origin, and $\mathbf{e}_x$ is the unit vector along the x axis.

Substituting (9) into (8) gives

$$H_{mn}(\varphi) = -\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iiint_{\mathbb{R}^3} p_{c,0}^*(\mathbf{r}_v - x_m \mathbf{e}_x, \omega_c) \times p_{s,0}(\mathbf{r}_v - x_n \mathbf{e}_x, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_v} d^3 \mathbf{r}_v. \quad (10)$$

Applying the change of variables $\mathbf{u} = \mathbf{r}_v - x_m \mathbf{e}_x$ gives

$$H_{mn}(\varphi) = e^{-ik_a x_m \sin \varphi} \left[-\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iiint_{\mathbb{R}^3} p_{c,0}^*(\mathbf{u}, \omega_c) \times p_{s,0}(\mathbf{u} - (x_n - x_m) \mathbf{e}_x, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{u}} d^3 \mathbf{u} \right]. \quad (11)$$

Since $x_n - x_m = (n-m)d$, define the displacement kernel

$$K_l(\varphi) = -\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iiint_{\mathbb{R}^3} p_{c,0}^*(\mathbf{r}_v, \omega_c) \times p_{s,0}(\mathbf{r}_v - l d \mathbf{e}_x, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_v} d^3 \mathbf{r}_v, \quad (12)$$

where $l = n - m$. Equation (11) then reduces to

$$H_{mn}(\varphi) = e^{-ik_a x_m \sin \varphi} K_{n-m}(\varphi). \quad (13)$$

The displacement kernel $K_l(\varphi)$ is the audio directivity generated by a reference carrier channel interacting with a sideband channel displaced by ld . It contains the element radiation, ultrasonic propagation, spatial overlap, and nonlinear interaction associated with that relative displacement. In contrast, the exponential factor in (13) accounts only for the absolute position of the channel pair.

Let $\Phi(\varphi)$ be the diagonal phase matrix whose m th diagonal entry is $[\Phi(\varphi)]_{mm} = e^{-ik_a x_m \sin \varphi}$. Equation (13) can then be written as

$$\mathbf{H}(\varphi) = \Phi(\varphi)\mathbf{T}(\varphi), \quad (14)$$

where the Toeplitz matrix reads

$$\mathbf{T}(\varphi) = \begin{bmatrix} K_0(\varphi) & K_1(\varphi) & \cdots & K_{N-1}(\varphi) \\ K_{-1}(\varphi) & K_0(\varphi) & \cdots & K_{N-2}(\varphi) \\ \vdots & \vdots & \ddots & \vdots \\ K_{-(N-1)}(\varphi) & K_{-(N-2)}(\varphi) & \cdots & K_0(\varphi) \end{bmatrix}. \quad (15)$$

Therefore, the complete N^2 channel-pair responses can be constructed from only $2N - 1$ displacement kernels, $K_l(\varphi)$ for $l = -(N - 1), \dots, N - 1$.

2.3. Mirror symmetry and banded approximation

Let $\mathbf{R} = \text{diag}(-1, 1, 1)$ denote reflection about the yOz plane. If the reference channel and propagation environment are mirror-symmetric, then $p_{\nu,0}(\mathbf{R}\mathbf{r}_\nu, \omega_\nu) = p_{\nu,0}(\mathbf{r}_\nu, \omega_\nu)$. Moreover, $\hat{\mathbf{s}}(-\varphi) = \mathbf{R}\hat{\mathbf{s}}(\varphi)$. Applying the transformation $\mathbf{r}_\nu \leftarrow \mathbf{R}\mathbf{r}_\nu$ to (12) yields

$$K_{-l}(\varphi) = K_l(-\varphi), \quad l = 1, \dots, N - 1. \quad (16)$$

Thus, the negative-displacement kernels can be reconstructed from the N nonnegative-displacement kernels. Equation (16) is an angular mirror relation rather than a complex-conjugate symmetry.

The integrand in (12) contains the spatial overlap between two ultrasonic fields separated by $|l|d$. This overlap is expected to decrease with channel separation, although its actual decay depends on the array geometry and must be verified numerically. Motivated by this behavior, define the banded approximation

$$[\mathbf{H}_L(\varphi)]_{mn} = \begin{cases} H_{mn}(\varphi), & |n - m| \leq L, \\ 0, & |n - m| > L. \end{cases} \quad (17)$$

The corresponding audio directivity is approximated by

$$D_a(\varphi, \omega_a) \approx \mathbf{w}_c^H \mathbf{H}_L(\varphi) \mathbf{w}_s. \quad (18)$$

Translation invariance, mirror symmetry, and banded truncation therefore reduce the number of independently evaluated channel-pair responses according to

$$N^2 \longrightarrow 2N - 1 \longrightarrow N \longrightarrow L + 1. \quad (19)$$

Algorithm 1 Alternating joint directivity control

Require: $\{\mathbf{H}_L(\varphi_i), d(\varphi_i), \alpha_i\}_{i=1}^M, \rho$

1: Initialize $\mathbf{w}_c^{(0)}, \mathbf{w}_s^{(0)}$, and $t = 0$

2: **repeat**

3: Form the sideband subproblem and compute $\lambda_s^{(t)}$

4: $\mathbf{w}_s^{(t+1)} \leftarrow \arg \min_{\mathbf{w}_s} [J_{\text{PM}}(\mathbf{w}_c^{(t)}, \mathbf{w}_s) + \lambda_s^{(t)} \|\mathbf{w}_s\|_2^2]$

5: Form the carrier subproblem and compute $\lambda_c^{(t)}$

6: $\mathbf{w}_c^{(t+1)} \leftarrow \arg \min_{\mathbf{w}_c} [J_{\text{PM}}(\mathbf{w}_c, \mathbf{w}_s^{(t+1)}) + \lambda_c^{(t)} \|\mathbf{w}_c\|_2^2]$

7: $t \leftarrow t + 1$

8: **until** the stopping criterion is met

9: **return** $\mathbf{w}_c^{(t)}, \mathbf{w}_s^{(t)}$

3. JOINT DIRECTIVITY CONTROL

Given the desired complex directivity $d(\varphi_i)$ at M observation angles, define the weighted pressure-matching error

$$J_{\text{PM}}(\mathbf{w}_c, \mathbf{w}_s) = \sum_{i=1}^M \alpha_i |\mathbf{w}_c^H \mathbf{H}_L(\varphi_i) \mathbf{w}_s - d(\varphi_i)|^2. \quad (20)$$

Here, $\alpha_i > 0$ is the angular weighting coefficient at φ_i , with a larger value assigning greater importance to the matching error at that angle. Although jointly nonconvex, the response is linear in $\mathbf{w}_s$ or $\mathbf{w}_c^*$ when the other vector is fixed. Algorithm 1 therefore alternates closed-form regularized least-squares updates with loading $\lambda = \rho \text{tr}(\mathbf{A}^H \mathbf{W} \mathbf{A})/N$, where $\mathbf{A}$ is the current subproblem matrix, $\mathbf{W} = \text{diag}(\alpha_1, \dots, \alpha_M)$, and $\rho > 0$ is the relative loading factor.

4. NUMERICAL RESULTS

4.1. Simulation setup

A 24×24 PAL is placed in the xOy plane and centered at the origin. The center of the transducer in row i and column j is $\mathbf{r}_{i,j} = [(j - 12.5)d, (i - 12.5)d, 0]^T$, where $i, j = 1, \dots, 24$. Each circular transducer has radius 5 mm, and the grid spacing is $d = 10$ mm, giving a closely packed, non-staggered array. All transducers in the same column share one driving signal, so the 24 columns constitute $N = 24$ channels. The carrier and upper-sideband frequencies are 40 and 41 kHz, respectively, producing a 1 kHz audio signal.

Directivity is sampled over $\varphi \in [-80^\circ, 80^\circ]$ at 1° intervals. The virtual-source domain spans $x \in [-15, 15]$ m, $y \in [-0.7, 0.7]$ m, and $z \in [0, 24]$ m. The finite domain is chosen to contain the translated virtual-source distributions of all channel pairs; otherwise, boundary truncation would violate the translation invariance assumed in the theoretical formulation. Both full and banded matrices are divided by the complex conventional-steering response at $\varphi_0 = 20^\circ$.

The target shown in Fig. 3(a) is obtained by multiplying the conventional complex directivity by a window centered at

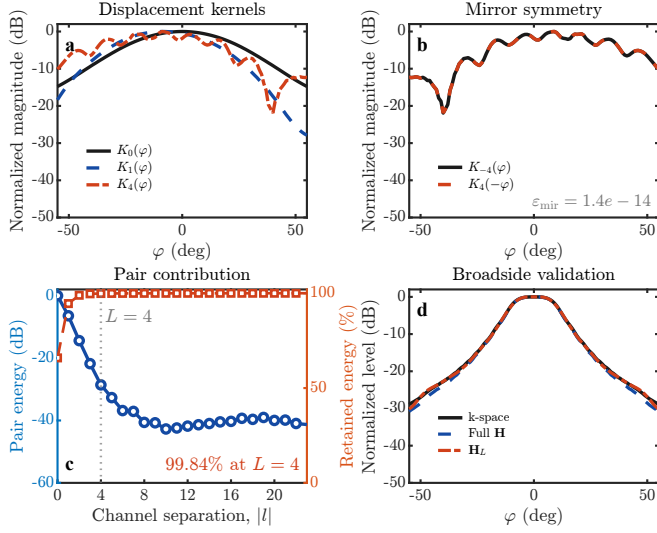


Fig. 2. Structured-model validation: (a) displacement kernels; (b) mirror reconstruction; (c) pair energy and cumulative retained energy; (d) broadside directivities from k -space simulation and the full and banded models.

φ_0 . The window is unity within $\pm 8^\circ$ and tapers to zero at $\pm 20^\circ$ through a raised cosine. The resulting target is normalized to $d(\varphi_0) = 1$. The angular weights are 20 within $\pm 8^\circ$, 10 where $|\varphi - \varphi_0| \geq 20^\circ$ and $|\varphi| \leq 55^\circ$, and 1 elsewhere. Joint PM starts from conventional steering weights; sideband-only PM fixes the same carrier weights and optimizes the sideband using the banded matrix. The loading factor is $\rho = 0.3$.

4.2. Structured-model validation

Figure 2(a) shows separation-dependent kernel responses. Figure 2(b) compares independently computed negative-displacement kernels with their mirror reconstructions. The relative complex errors, $\|K_{-l}(\varphi_i) - K_l(-\varphi_i)\|_2 / \|K_{-l}(\varphi_i)\|_2$, are 3.89×10^{-15} and 1.41×10^{-14} for $l = 1$ and $l = 4$, respectively, with norms over all sampled angles.

For $\mathcal{P}_l = \{(m, n) : |n - m| = l\}$, the mean pair energy is $E_l = |\mathcal{P}_l|^{-1} \sum_{(m,n) \in \mathcal{P}_l} \sum_{i=1}^M |H_{mn}(\varphi_i)|^2$. The cumulative retained energy is $\eta_L = \sum_{l=0}^L |\mathcal{P}_l| E_l / \sum_{l=0}^{N-1} |\mathcal{P}_l| E_l$.

In Fig. 2(c), $L = 4$ retains 99.84% of the matrix energy, requiring only $L + 1 = 5$ independent kernels instead of $N^2 = 576$ pair responses without the proposed optimization. Figure 2(d) shows close agreement among direct k -space simulation, the full model, and the banded model under uniform broadside excitation.

4.3. Joint directivity control

All responses are evaluated using the full matrix and normalized by their respective peak magnitudes. Performance

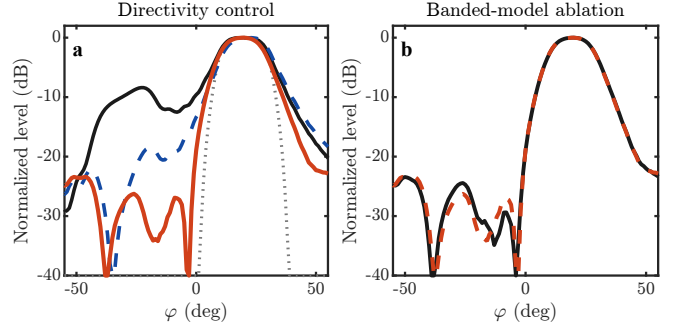


Fig. 3. Directivity control at 20° . (a) target, — conventional steering, -- sideband-only PM, and — joint PM. (b) Joint PM optimized using the — full and -- banded models; both responses are evaluated using the full matrix.

is measured by the weighted target error $\varepsilon = \|\mathbf{W}^{1/2}(\mathbf{D} - \mathbf{d})\|_2 / \|\mathbf{W}^{1/2} \mathbf{d}\|_2$ and the maximum level over $|\varphi - \varphi_0| \geq 20^\circ$, $|\varphi| \leq 55^\circ$, relative to the main-lobe maximum within $|\varphi - \varphi_0| \leq 15^\circ$.

Conventional steering applies frequency-dependent phase shifts to both ultrasonic components to steer the carrier and sideband beams toward φ_0 . Sideband-only PM fixes the carrier at its conventional steering weights and optimizes only the sideband weights, whereas joint PM optimizes both weight vectors alternately.

In Fig. 3(a), conventional steering, sideband-only PM, and joint PM yield suppression levels of -8.4 , -10.1 , and -14.1 dB and target errors of 0.374, 0.210, and 0.125, respectively. Figure 3(b) shows that full- and banded-model optimization achieve similar suppression levels (-14.1 and -14.1 dB) and target errors (0.123 and 0.125), confirming that the five-kernel model preserves the full-model control performance.

5. CONCLUSION

This paper developed a structured bilinear model for multi-channel PAL directivity control. Each channel-pair response is separated into an absolute-position phase and a relative-displacement kernel. Translation invariance and mirror symmetry reduce the number of independent responses from N^2 to N , while banded truncation further reduces it to $L + 1$. For the considered 24-channel PAL, five kernels retain 99.84% of the matrix energy and closely reproduce the full-model directivity and control performance. Joint carrier-sideband PM reduces the maximum suppression-region level by 5.7 dB over conventional steering while maintaining the desired beam direction. Thus, exploiting the effective sparsity of nonlinear channel interactions reduces model construction while preserving control performance.

6. COMPLIANCE WITH ETHICAL STANDARDS

No ethical approval was required for this study. The authors have no relevant financial or non-financial interests to disclose.

7. ACKNOWLEDGMENT

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